HISTORICAL AND TECHNICAL RETROSPECTIVE

Reverse engineering beads photo 2

The development process, final evidence and lessons for future projects

Through R347 and its maker answers, with the R348 analysis.

Faithful PDF export of photo2/RETROSPECTIVE.md
Source revision: 988402b6
Figures are embedded; evidence links open immutable GitHub sources.

---

1 Executive summary

2 Repository access and branch inventory

3 Chronological history

4 Technical analysis

5 Why the investigation took so long

6 What the final state establishes

7 Lessons and practical alternatives in hindsight

Evidence links

Reverse engineering the necklace in beads photo 2

Historical and technical retrospective through R347 and its maker answers, updated in R348, 20 October 2026. Reconstruction baseline: `8e9648b9`; R347 documentation/evidence delivery: `5d605663`, `photo-2-strategy-v3`. R348 adds documentation and saved-graph sensitivity analysis only. Historical scientific products remain unchanged. Practical recommendations and a reusable failure-detection checklist are in Future project methods.

1 Executive summary

The project has produced a strong candidate for the necklace's original color pattern, a substantial photographic lattice of bead relationships, a complete executable POV-Ray reconstruction, and a much better approximation to the necklace's broad cast shadow. **The result is close to the photograph, but the exact original stringing pattern and physical reconstruction have not been conclusively recovered.** That distinction agrees with the maker's current assessment.

Current status through R347: the unchanged 208-position candidate has **201 supported, 0 conflicting and 7 insufficiently corroborated positions**, with all 208 positions observed and **1,040/1,040 retained color observations agreeing** under the current indexing. Two old red readings were corrected to yellow by maker inspection of the original photograph. A third, R332-R-0037-0040, is withheld from pigment evidence at the maker's request because the photo is insufficient; its selected yellow depended on consulting the reconstruction and supplies no independent confirmation. The seven insufficient positions remain provisionally black. Raw answers and assessment

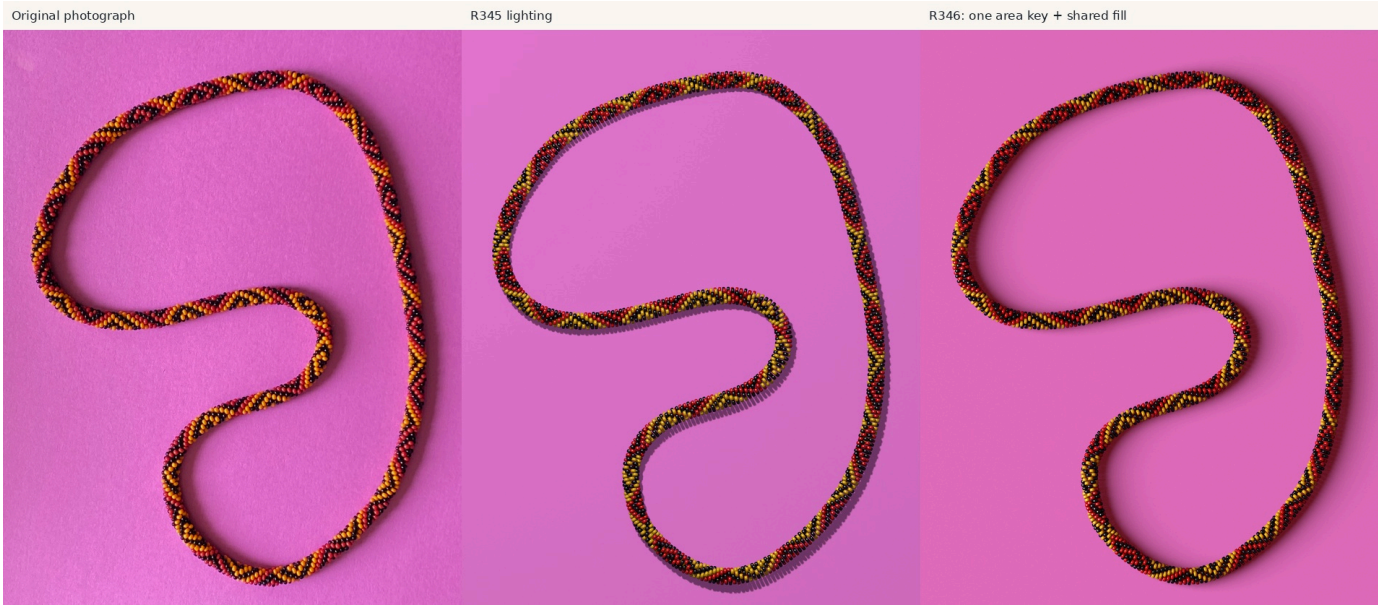
The R347 divisor calculation leaves **A/208 as the unique compatible case within the fixed lattice, H=(214,201), two specified family/count conventions, complete-repeat and shortest-period-below-400 assumptions**. R348 additionally finds that deleting the questionable node and its three links from a derived graph leaves connectivity, the cycle-displacement generator H, all remaining coordinates modulo H, pattern coverage and this conditional result unchanged. Neither calculation independently verifies the physical necklace, original chart, bead count, exact topology or geometry. Conditional certificate, deletion sensitivity and limits

Historical R344 remains 199/2/7 and 1,038/1,041, with three contrary software-red records at positions 83 and 189. Those are the correct figures for that evidence epoch, not the current unresolved-conflict count. R347's corrections and exclusion are saved separately; the provisional sequence and historical graph are unchanged. R344 report, historical pattern with evidence

This is much stronger than a pleasing render or a repetition guessed from a few similar patches. Geographic prediction, whole-occurrence prediction, comparison with shorter and alternative patterns, exact lattice equations, and direct photographic review support the candidate. Nevertheless, most labels are software or analyst interpretations of the same repeatedly studied photograph. The topology is a working reconstruction with preserved alternatives. Agreement with those labels is not independently measured accuracy against the original necklace.

R345 constructs **2,704 separate rounded, hollow beads**, repeating the provisional 208 block thirteen times. Its selected geometry closes and has separation certificates for all 8,112 nearby pairs requiring explicit examination. This resolves the actual interpenetration problem for that rendered scene. It does not establish that the necklace really contains 2,704 beads, that every photographic support has the right physical identity, or that its camera and centerline are measured correctly. Those remain conditional or approximate inputs. R345 report, geometry validation

R346 is a genuine final advance in illumination. It assesses **the cast shadow of the whole necklace on the paper**, on both sides around the loop, rather than fitting selected bead shadows or specular spots. One fixed broad area source and a shared warm room contribution reduce paper-shadow profile RMS error from **20.4 to 6.8 percentage points**, and held-station error from **20.0 to 6.4**. Same-coordinate paper comparisons also improve. The reconstructed global shadow is convincingly closer. Individual bead highlights, local shape differences, and paper texture remain visibly imperfect. The fitted light is a plausible arrangement, not a unique recovered lamp calibration. R346 report, saved measurements



The final comparison supports substantial visual similarity. Enlarged views still reveal a more regular, thinner, glossier bead arrangement and different background texture. The broad shadow improvement is visible independently of those differences.

The length of the investigation has several causes, with different evidential strength:

1. **The inverse problem is genuinely ambiguous.** Hidden beads, repeated colors, occlusion, uncalibrated scale and camera, and uncertain surface ownership couple discrete topology to continuous geometry and illumination. A highlight is not a bead center, and a color mask is not a bead identity.
2. **Several methods optimized insufficient or misleading targets.** Early image-region order was substituted for crochet order; center proximity rewarded excessive bead density; brightness extrema and smooth bands were treated as stronger identity cues than they warranted; later objectives improved highlights while degrading visible material. These are documented computational problems, not unavoidable consequences of using one photograph.

3. **Search failure was real, but not a universal explanation.** R301 failed to recover known synthetic geometry even though the accurate forward model represented the exact answer. R311 found missed basins. Conversely, broader searches often traded one photographic error for another, and R319 did not establish a true nonlinear local minimum or prove that its model family was incapable.
4. **The process repeatedly worked on difficult prerequisites longer than necessary.** Useful manual observations, graph algebra and incomplete-word periodicity were available relatively early. Much effort went into increasingly elaborate segmentation and optical fitting before a robust photographic lattice and simple pigment audit were in place. Later work made especially productive use of those simpler constraints.
5. **Preservation improved more than synthesis at first.** The late history is unusually reproducible and candid about failure. However, old discoveries were sometimes preserved without remaining effective constraints, and measurements, labels and approval scopes were occasionally communicated too strongly. R297, R329, R332, R336 and R343 provide concrete examples.

Branch changes generally reflected the maker's dissatisfaction with the preceding work, as the maker clarified during this retrospective. They were not all independent projects or successful milestones. The Git relationships show which changes were true continuations and which were deliberate restarts.

2 Repository access and branch inventory

Access and preservation

All three repositories were present locally before substantive investigation:

Repository	Local clone	Origin
beads	`/home/rharris/git/beads`	`git@github.com:rharriszzz/beads.git`
bead_map	`/home/rharris/git/bead_map`	`git@github.com:rharriszzz/bead_map.git`
hsv_tools	`/home/rharris/git/hsv_tools`	`git@github.com:rharriszzz/hsv_tools.git`

All origins were fetched with `--all --no-recurse-submodules --no-auto-gc`. Each uses the fetch refspec `+refs/heads/*:refs/remotes/origin/*`. All clones are non-shallow; an object traversal found no missing objects in their locally reachable histories. No clone or large binary download was needed. Fetching the two sibling repositories required access outside the read-only filesystem sandbox; those fetches succeeded. A sandbox DNS failure during an additional beads reference check also resolved with the permitted unrestricted Git command. There was no demonstrated GitHub authentication problem.

The original retrospective began on `photo-2-strategy-v3`, matching origin at `8e9648b9`. Existing untracked R340 work in beads and a modified `hsv_picker.py` in hsv_tools were preserved. Historical analysis of hsv_tools uses `git show HEAD:...`, not that modified working file. That initial investigation performed no checkout, source edit or historical scientific rerun. The later authorized documentation/R347 delivery was committed and pushed as `5d605663`; R348 extends it on the same branch, preserving unrelated work and historical scientific outputs.

The machine-readable branch inventory, commit index, and access record preserve the original pre-R347 snapshot: **416 reachable distinct commits in beads, 5 in bead_map and 10 in hsv_tools**. These are repository-history counts at that snapshot, not current totals or counts of independent experiments. Numbered R entries also include questions, clarifications, reviews and delivery steps.

All branches newest first

The following table preserves the original through-R346 branch snapshot. Dates are tip **committer dates**, retaining their recorded UTC offsets. Local and matching remote-tracking references are one branch, not two developments; `origin/HEAD` is a symbolic default-branch reference and is not an extra branch. The subsequent v3 tip before R348 is `5d605663`, dated 2026-10-20 19:49:18 −04:00, adding the retrospective and R347 evidence rather than a new reconstruction branch.

Repository and branch	Latest commit date	Tip	Purpose and relationship
beads `photo-2-strategy-v3`	2026-10-20 16:44:04 −04:00	`8e9648b9`	Latest reconstruction strategy, R233–R346; descends directly from reconstruction-v2. Photographic evidence, physical fitting, lattice recovery, pattern audit, final rendering and whole-necklace shadow.
beads `master`	2026-10-15 17:31:55 −04:00	`5d1258cd`	Original forward-rendering/photo repository plus sine/radius documentation. Its recent date does not make it the latest inverse reconstruction.
beads `photo-2-reconstruction-v2`	2026-10-05 19:03:11 −04:00	`b28b99e8`	Restart from the older master base, R092–R232. Background, local geometry, manual graph evidence, distributed supports and seed-selection work.
beads `photo-2-reconstruction`	2026-09-27 15:10:37 −04:00	`2c4c116b`	First photo-2 reconstruction, R001–R091, based on image-to-pattern-2; initial rendering, synthetic indexing checks and reviewed generated-image inventories.
beads `archive/image-to-pattern-2-wip`	2026-09-23 11:10:21 −04:00	`debec405`	One archival commit beyond image-to-pattern-2. Preserves unfinished uncommitted v2 work and an earlier render; not a new successful reconstruction.

Repository and branch	Latest commit date	Tip	Purpose and relationship
beads `image-to-pattern-2`	2026-06-09 17:46:24 −04:00	`402663eb`	One plan-replacement commit beyond image-to-pattern. A methodological restart plan, not a completed second pipeline.
beads `image-to-pattern`	2026-01-05 17:31:43 −05:00	`26ba79a9`	January generic inverse pipeline, layout tests, segmentation/palette experiments, manual color tooling and saturation/FFT diagnostics.
bead_map `main`	2025-04-27 21:25:19 −04:00	`46e17d63`	Applies manually produced HSV masks to bead detection, nearest-neighbor ordering and skeleton/spline extraction. Separate Git root; directly dependent on hsv_tools outputs.
hsv_tools `main`	2025-04-26 15:22:44 −04:00	`3e70fe30`	Interactive HSV exploration, histogram/bucket counting, line-based selection, merged masks and saved photo-2 masks. Separate Git root.
beads `pattern`	2025-04-24 16:42:43 −04:00	`84672643`	April segmentation/quantization/periodicity prototype branched from the photo-file rename commit; not an ancestor of the later inverse branches. Available locally as `origin/pattern`, without a separate local branch.

Actual ancestry

The important split is at `020303ec`. Reconstruction-v2 is **not** a descendant of the first photo-2 reconstruction. It starts afresh from their older common master base and carries selected earlier knowledge through documentation and copied evidence. The final v3 branch then adds 194 commits beyond v2.

```
c100ac8c  original forward scene and website, 2017
  └─ 895f8aa7 — 5bbd0e0f  photographs and renaming, April 2025
                  ├─ pattern → 84672643
                  └─ 020303ec  more photographs
                       ├─ master → 5d1258cd  sine note
                       ├─ image-to-pattern → 26ba79a9
                       │    └─ image-to-pattern-2 → 402663eb
                       │         ├─ archive/image-to-pattern-2-wip → debec405
                       │         └─ photo-2-reconstruction → 2c4c116b
                       └─ photo-2-reconstruction-v2 → b28b99e8
                            └─ photo-2-strategy-v3 → 8e9648b9

hsv_tools and bead_map have separate roots, connected by shared mask files.
```

The sine note on master is the same file blob as the note introduced by `81234cba` on v3. The different commit hashes and separate ancestry do not imply two independent discoveries or a merge of the scientific branches.

What each line left behind

Branch or repository	What remained unresolved at its stopping point	What survived
hsv_tools	Pixel selections did not determine distinct beads, physical centers, topology or a repeat. No validated full-pattern output is stored.	Human-guided color sampling, circular hue comparisons, saved masks and an interactive way to inspect source pixels.
bead_map	Mask polarity/scale handling and shadow contamination were fragile; greedy proximity did not recover crochet order. No established photo-2 sequence was delivered.	Recognition that centerline, scale and ordering needed explicit treatment; early debugging plots and the exact masks linking both repositories.
pattern	Segments and palette clusters were not established bead identities, and region order did not establish string order.	The goal of reversing beads.pov, the 6/7 neighborhood idea, and separation of segmentation, color assignment and periodicity. Later branches did not inherit this implementation through Git.
image-to-pattern	Real-image and rendered-JPEG recovery remained poor despite passing component tests. Palette/background mixing, sparse retention and ordering remained unreliable.	Forward-layout tests, source-pattern fixtures, geometry helpers, manual color configuration, and an unusually useful failure record in its plan.
image-to-pattern-2	The new milestone plan had not been completed.	Explicit lessons about per-stage validation, known-pattern photo 4, partial observations and avoiding an untrusted end-to-end baseline.
archive branch	Unfinished source/renderer-parity work, not a solved inverse pipeline.	An accessible record of otherwise uncommitted work and the prior render.

Branch or repository	What remained unresolved at its stopping point	What survived
photo-2-reconstruction	No complete pattern from any of seven generated JPEGs; automatic image-neighbor indexing and physical photo geometry unvalidated.	Original image-axis spline, supplied-edge integer propagation, missing-color periodicity, shape/visibility distinctions and documented failed cues.
reconstruction-v2	No accepted global count, hand, full bead correspondence or robust broad masks. The tip retained M4 as a provisional seed approach, with misses and a same-body question.	Maker graph labels, confirmed positive interiors, distributed color observations, exact surface/visibility tools and the spline reference later reused by R345.
strategy-v3	Exact physical count/topology, nine pattern entries, full geometric correspondence and unique camera/optics remain uncertain.	A connected working lattice, strong conditional 208 motif, conservative pigment corrections, a nonintersecting full scene and a much improved global cast shadow.
master	No inverse reconstruction on this branch.	Original bead macro, example patterns and documented radius conventions.

3 Chronological history

2017 to April 2025 The forward model and the first inverse attempts

The initial commit, `c100ac8c`, already contains a functioning conceptual forward model: a repeated linear color array placed around a minor helix whose axis follows a major circle. The bead macro models rounded hollow glass rather than colored points. This was valuable prior knowledge, but it was an illustrative construction model, not a measured mechanical model of photo 2. The photography was added in April 2025.

The `pattern` branch tried SLIC superpixels, region-adjacency merging, palette clustering and repetition detection. Its documentation describes an iterative segmentation/quantization approach, but the code exposes a larger problem than color noise. In `main.py`, the initial color sequence is extracted by sorting region records; that ordering is not physical crochet order. The optional graph path between outer and inner background regions is also not the complete string traversal of a tube. `pattern.py` attempts three distance clusters corresponding to the 1/6/7 neighbor families, but image distance alone does not identify those families around bends and occlusion. Prototype description, main program, ordering implementation

This was a reasonable exploratory decomposition, but the ordering step lacked the physical constraints needed to make an autocorrelation or exact string-period result meaningful. The saved code and documentation do not establish successful recovery of photo 2.

The other two repositories belong to this same attempt. `hsv_tools` developed histogram/bucket exploration and a two-click line picker that collects HSV values, with circular hue tolerance and an option to merge selections. Its final commit saves red, yellow, black and background masks for `beads-photo-2.jpg`. `bead_map/bracelet-pattern-extractor.py` explicitly says these masks were made by hand

with hsv_tools. The red, yellow and black mask blobs are **identical in both repositories**, which establishes an actual artifact transfer rather than merely similar intentions. HSV picker, HSV counts, mask consumer

`bead_map.py` estimates an apparent radius from component areas, proposes LoG blobs, constructs a KD-tree and follows a nearest-neighbor cycle. Its estimate of total count is based on detected visible candidates, so it cannot independently count hidden beads. The spline helper in `bracelet-pattern-extractor.py` closes masks, skeletonizes them, chooses widely separated skeleton points and routes between them. This is an initial geometric heuristic, not a reliable complete closed necklace reconstruction. bead_map implementation

There are also identifiable implementation hazards: mask polarity is treated differently between color classes in `bead_map.py`, and the spline helper compares `imread(..., as_gray=True)` values with 128 even though those grayscale values use a normalized float scale. These are static code findings; the history does not isolate their contribution to a particular saved run. More fundamentally, even corrected masks and blob coordinates would not make a nearest-neighbor walk equal to the crocheted string's order. The June plan later explicitly describes the sibling centerline code as unreliable around cast shadows. June plan

January 2026 A modular pipeline without a trustworthy full recovery

`image-to-pattern` implemented layout, source patterns, segmentation, sampling, palette inference, bead detection, graph ordering and periodicity as tested modules. This was more organized than the April prototype. It also accumulated a GUI for manual color rectangles, HSV mask tools, sparse-mask heuristics, saturation segmentation, unwrapping and FFT diagnostics. January plan and results, pipeline code

The tests often exercised simplified bands, grids or supplied layout information. The plan's real-image results are more informative than the test count. Early auto-geometry produced only three samples on a generated bracelet because multiple passes inflated measured width. A longest-run correction helped that defect. Nevertheless, CLI examples still returned a one-color period: photo 2 had approximately 159 samples and pattern `[0]` in that recorded trial. K-means mixed magenta paper into inferred colors and split dominant hues. Later fixture comparisons sometimes improved, but often retained very few samples, so higher best-case matches did not establish a complete sequence.

The early plan even described photo 2 as suggesting a purple/skin-tone palette; the magenta was paper. This is a concrete example of scene interpretation being wrong before downstream inference began. Manual color tooling was a productive response. GUI refinements and many mask-selection iterations were usable infrastructure, but they did not resolve the relationship between visible regions and physical string indices.

The branch stopped with a saturation-based spline/unwrap/FFT debug script on 5 January. There is no demonstrated end-to-end recovered photo-2 pattern at that tip.

June and September 2026 A new plan and an archive

The sole new commit on `image-to-pattern-2`, `402663eb`, replaces the plan. It explicitly identifies the earlier lack of a trusted baseline, late per-stage metrics, and movement to difficult images before an easier image had been solved. It points to **photo 4 and POV case 4** as a known real-photo pattern

pair, and recognizes hsv_tools and bead_map as preceding work. These were valuable methodological corrections. They were a plan, not evidence that those milestones had passed. Plan v2

The September archival branch preserves an old render and the unfinished `pattern_from_photo` source/parity work in one commit. The archival date does not establish when every previously uncommitted file was written. It should not be counted as a separate successful experiment.

September 23 to 27 R001 to R091 Establishing what the measurements mean

The first `photo-2-reconstruction` branch generated an initial full-necklace scene around a saved image-derived spline. Its count, camera, materials and pattern remained provisional. The working count of 2,698 was a model hypothesis, not a physical census. The source centerline had 303 points and came from earlier image-only boundary work; later analyses repeatedly reused it. Branch overview, methods at R091

This branch made important conceptual progress. Maker guidance supplied the **±1, ±6, ±7** neighbor steps. R030/R031 established exact index propagation when correct signed edges were supplied. It also demonstrated that a wrong isolated bridge can be internally cycle-consistent. R036 tested partial-word repetition with missing colors while preserving original positions. This avoided the serious error of compressing away hidden or unreadable beads. These combinatorial components were viable well before the final pattern was recovered. Neighbor audit, sequence audit

Finding the image edges was much harder. Centroid/outline cues improved some synthetic pair precision, but signed-label errors left the tested full graphs inconsistent. Joint triangle/trace constraints rejected false bridges while losing many true edges. Local contour fitting could recover favorable synthetic patches with supplied camera/rope geometry, but did not transfer reliably or settle helicity. Conservative highlight repair recovered omitted pixels; more aggressive shared-cavity assignment added wrong-owner pixels. Method comparison, joint inference, local fitting, highlight comparison

The task then concentrated on blind, image-only inventories of `beads1.jpg` through `beads7.jpg`. Reviewed maps improved markers, colors and sliver handling, including black/neutral cases. However, none yielded a complete recovered pattern. This was useful evidence about visibility and measurement failures, but it became a substantial detour from the target photograph. R092 explicitly returned priority to photo 2. The first branch's ending product was a clearer account of approximate masks, boundaries, centerline and candidate observations, rather than recovered physical identities. R091 methods, lessons carried forward

The new branch did not simply erase those results. It retained the spline, local shape reasoning and the distinction between visible centroids, physical centers and highlights. Their numerical pipeline and reviewed masks were deliberately not imported wholesale.

September 27 to October 5 R092 to R232 Local evidence improved but segmentation remained a bottleneck

Reconstruction-v2 restarted from the older master base. Its opening work compared paper/background methods: local Gaussian FFT texture, spatial texture, learned paper appearance, manual labels and smooth contours. Shadowed paper could overlap bead appearance in HSV, so a perfect global hue cutoff was not available. A major practical question was which boundary errors could be bridged from reliable neighboring geometry. Background methods, parallel paths

R114 began with a small assisted patch. Maker feedback separated bodies that proximity alone had confused. R144–R147 extracted a useful integer graph from manual numbered direction series: 27 visible beads and 54 links, with all 28 independent cycles closing after two skipped-step entries were corrected. The correction was concrete: `10→12` became `10→11→12`, and another path was expanded through omitted intermediate names. This showed how small, explicit human annotations plus exact consistency checks could remove ambiguity cheaply. Numeric maker bead names were correctly kept separate from physical string indices. Maker series analysis

Positive interior constraints were also useful, but did not uniquely determine geometry. R160 refitted both family interpretations and found models of both hands covering all confirmed interior samples and all maker marks, including a fixed-6.5 check. An earlier apparent advantage for one saved pose disappeared when the competing family was actually refitted. This is a particularly clear warning against interpreting a local optimizer's failed start as physical disproof of a family. Both-family refit

R175–R190 extended provisional whole-necklace models, varied count, displayed exposed minor-outward points, collected 41 distributed maker marks, and attempted a bead-derived centerline. The count change from 2,698 to 2,833 was a requested 5% hypothesis, not a recovered total. The independent candidate centerline did not improve the saved-center reference check; synthetic scenes exposed bias from the asymmetric distribution of visible colored interiors. The old spline survived because the replacement lacked stronger evidence, not because it had become a measured physical axis. Tangent-circle model, centerline comparison

R192–R214 built trusted positive interiors and attempted registration using those pixels and maker marks. Close same-color points were sometimes two beads despite a bright connecting bridge; one correctly placed seed could still grow into another body. R214 showed that center distances and patch containment favored different count hypotheses: denser models could place some center near every observation while explaining less of each real surface. Same-body constraints were more informative than raw nearest distance. Ownership audit, registration

R215–R232 returned to broad colored segmentation and seed selection. Diffuse-core refinements improved rendered-fixture recall but produced real-photo masks that the maker rejected: one included black material, another spilled into adjacent yellow. The four-method seed comparison at R231 was a more useful causal comparison than another aggregate score. M4's local profile support looked most promising to the maker, but retained misses. Crucially, **the rejected old seeds also passed ownership on the synthetic controls**. Those controls therefore failed to reproduce the real failure and could not validate its repair. Mask failures, seed methods, maker's seed review

The branch stopped at that provisional seed review. The dissatisfaction was not just that some edge pixels were inaccurate. A reliable one-observation-per-bead inventory, global index map and recovered pattern were still absent after many rounds of tools and local improvements.

October 10 to 11 R233 to R254 A new strategy and a bounded global test

V3 continued directly from v2, rather than creating a fourth independent numerical base. The Stage 1 pilot, recovery and ownership work made the precision/coverage tradeoff explicit. R239 removed confirmed false, duplicate and wrong-color cases in six audited sections, but coverage fell: familiar-section support changed from 103/109 to 96/109; fresh-section support fell from 41/44 to 30/44. Nine

accepted proposals still had unresolved ownership. The report correctly concluded that a useful subset could support a bounded correspondence experiment without perfect global segmentation. Stage 1 diagnosis, ownership result

FFT work then separated frequency-vector directions from spatial bead rows, explored Gaussian width/cutoff choices, compared V and luminance-like Y, retained competing trios, and tracked peaks around the necklace. Dense tracking showed rank changes, gaps and fragmentary continuity rather than three perfectly persistent physical tracks. R249 found stable candidate trios at two of six sites in V and four in Y; that did not itself assign physical 1/6/7 families or bead identities. FFT transfer, dense tracking

R252 consolidated **310 distinct red/yellow observations** from overlapping earlier evidence, with source alias checks and geographic coverage. This was an especially productive synthesis step: earlier counts were not simply added together, and ambiguous samples were withheld. Observation consolidation

R253 compared count/hand hypotheses using these observations. Its leading counts were around 2,670–2,700, with substantial systematic uncertainty; FFT integration suggested approximately 2,675–2,679. The best finite geometry fit did not establish an exact integer. R254 then tested exact repeat compatibility and bounded correspondence alternatives. Every tested admissible exact pattern conflicted with retained colors within its saved domains. The point was not that the necklace lacked a repeat, but that the tested model-to-photo mappings were insufficient. Global count comparison, repeat infeasibility audit

At this stage a hard instruction excluded shortest lengths divisible by 13. That restriction is historically important, but **it was later withdrawn in R337**. Treating it as a current physical fact would wrongly reject the final candidate 208.

October 11 to 14 R255 to R282 Native detection and the separation of image evidence from models

These rounds built increasingly inspectable photographic evidence: full-loop audits, surface debuggers, photo-only detection, native-pixel maxima/growth, reflection ownership, color stops and coverage checks. The progress was real, but candidate count repeatedly had to be distinguished from distinct bead count.

R261's maximum-catchment ownership made supports highly fragmented. R262 removed that watershed veto, allowing independently grown supports to overlap. Its motivating pair shared 259 of 261 pixels after the change yet remained two proposals, correctly exposing duplicate support without assuming two beads. R263 explored parameters against an approximate visible-count target; a high brightness cutoff reached a plausible count while discarding 92 of 97 baseline dark-enclosure supports. That is a direct example of why matching an expected count was an inadequate quality objective. Native growth, watershed removal, parameter study

R270 compared brightness definitions and handled the source's embedded Apple Wide Color Sharing profile. Luminance/OKLab found useful pale-yellow locations that max-channel V missed, but permissive spatial searches could still cross into the wrong bead. Better photometry helped coordinate extraction without solving ownership. Brightness comparison

R276 integrated literal rounded-annulus geometry, all-bead occlusion and shared lighting. It exposed extensive physical intersections in inherited placement. R278's corrected radius convention removed the sampled +1 collision issue but did not automatically solve diagonal packing or indexing. Fourteen new closed candidates had separation witnesses, yet none explained substantially all photographed bodies. This was a valuable distinction: **a physically valid candidate can still be the wrong reconstruction**. Integrated lighting/geometry, radius and correspondence

The sine error deserves precise treatment. Historical source used `sin(180/6.5)` with a radians-valued sine function. The intended half-angle is `pi/6.5`. This changed an important geometric scale. It was already recognized and deliberately preserved as a legacy convention in earlier parity work; R278 tested the corrected convention, the October 15 note explained it explicitly, and R345 made the corrected convention the default while retaining `LegacyRadius=1`. It is a significant modeling defect, but cannot explain all later failure: corrected, separated candidates still had photographic mismatches. Sine note, R278, R345

R282 implemented an end-to-end three-phase system: photographic measurements, physical correspondence/geometry, then pattern inference. Its first pass learned a background-tinted bead hue and produced misleading rankings; later passes repaired specific problems while preserving all epochs. The final result remained conditional. Integration made errors visible; it did not turn uncertain measurements into verified physical identities. R282 integrated report

October 14 to 15 R283 to R293 The limits of detailed segmentation

These rounds clarified a permanent measurement distinction. Colored supports were photographic interiors, not complete silhouettes. Black supports were reflection measurements, not inferred dark bodies. J/M/W and related methods compared isophotes, diffuse intensity, watershed competition, dark boundaries and local separators. They repeatedly encountered a fundamental ownership issue: a brightness contour can bypass a valley, split one bead, or join touching beads of the same color. Measurement contract, J and M explanation

R289 independently detected connected bright patches and found a simple approximately five-pixel displacement useful for predicting historical colored-interior centroids on a qualified subset. It explicitly did not calibrate physical centers. Multiple red highlights and synthetic transfer failures prevented treating this as a general geometric solution. Independent landmarks

R290–R292 improved review and separator handling, but R293 diagnosed a process problem: difficult historical cases and fresh-target exclusion rules had shifted review toward peripheral interfaces. Eight of eleven R292 targets lay beyond the stated near-centerline reach convention. They were useful regression cases, but a poor population for judging broad progress. R293 refocused on clear central evidence and allowed a bounded physical trial without another large contour rewrite. R293 assessment

October 15 to 18 R294 to R321 Extensive physical fitting without a final accepted geometry

This was the largest concentrated physical-model campaign. It should neither be dismissed as useless nor presented as a progressively solved inversion.

R294 fit a 221-address top section with exact annular visibility and contact checks. A better-looking unconstrained revision contained **93 demonstrated penetrating pairs**; the physically certified revision did not solve important central colored-surface mismatches. R295 alternated discrete ownership alternatives with continuous geometry and improved some reserved large-region matches while retaining failures. R294, R295

R296 extended fitting to the full necklace. R297's paired diagnosis established why an earlier useful local fit regressed: global normalization changed the local metric as total count changed, and closed winding changed phase progression. Restoring metric/phase recovered earlier local containment, but did not fix every nearby body. A preservation term then made the tradeoffs explicit. This is a direct example of previous progress being archived yet not initially protected by the extended parameterization. R297 causal diagnosis, paired native-support records

R298 measured shared bead proportions, transverse widths and the planar axis, exposing camera image-plane rotation as an important previously fixed degree of freedom. Thin-bead/wider-rope fits improved those selected dimensions but regressed separators, paper occupation and earlier fits; the primary was not adopted. R299/R300 calibrated and inspected shared-light reflection residuals around the loop. Their magnitudes did not simply accumulate, but signed geographic biases and uncertain joins prevented stable physical indexing claims. R298, R300

R301 compared Sobol/differential-evolution search with coordinate controls and coupled camera, shape and light fitting. It achieved some lower reflection errors, but worsened dimensions and visible material. More decisively, it failed synthetic known-geometry recovery even with known owners. The exact generator could reproduce the observations; the coarse predictor and finite search did not reliably find it. This supports an optimization diagnosis, not a claim that the real necklace was mathematically unreconstructable. R302's sensitivity-guided hierarchy improved the local comparison with fewer dimensional regressions, but still did not identify unique physical constants. R301 report, generating-parameter predictor check, recovered versus true parameters, search implementation, R302

R303 derived the actual source relationship between axial height and turn advance. R304/R305 extended a local model over 350 and then 1,041 consecutive addresses; visible errors fluctuated and earlier collisions remained. These were valuable stress tests of extrapolation, not successful global recovery. R303, R304, R305

R306–R309 executed full-circuit optimization and improved performance and some component scores. They did not establish substantial whole-photo improvement. R308 found an inherited false boundary reward involving unresolved rays. R310 showed nearby material failures before a verified physical chain existed; describing a late point as the first accumulated error was premature. R311 audited actual search breadth, found missed global basins and obtained partial repairs, but no finalist improved every important geographic/component measure. R306, R308, R310, R311, actual search coverage

R312–R319 developed minimax reflection objectives, geometry gates, better ray handling, layered rendering and explicit search-space audits. These were substantial numerical advances. They also revealed why their scalar scores could not be called a recovered scene:

- R314 corrected a tautological persistence expression and refused to begin highlight minimization when geometry failed its gates.

- R315 resolved small unknown-ray and censored-boundary errors; those numerical repairs did not explain the much larger coherent surface mismatches.
- R316/R317 substantially improved some early cores with shared geometry while leaving other photographic conditions unsatisfied.
- R318's fixed-map reflection minimax stalled around 12 pixels. R319 showed coupled conflicting responses, many fixed or inactive serialized parameters, noisy secants and real collisions under larger steps. It did **not** certify a nonlinear local minimum or global impossibility.

R314, R315, R316, R317, R318, R319 search audit

R320/R321 finally derived more of the staircase packing from count, dimensions, contact support and curvature. R321 repaired a photo-sized closed N2800 candidate's collisions without moving its centerline, for both hands. This established that one earlier collision failure arose from the spacing law, not unavoidable physical impossibility. It still did not establish photographic correspondence or a final scene. R320, R321

October 18 R322 to R329 Photographic topology became the main source of progress

R322 directly tracked photographic diagonal bands. It recovered most of the maker's interior test patch, including a difficult black-highlight sequence, but made three wrong interior links. Smoothness, stable bandwidth and cycle closure had not prevented a band from switching physical rows.

R323 compared **joint multi-support row alternatives across overlapping windows**. It recovered all 25 clearly interior reference beads and all 45 interior links, with zero wrong interior links in that bounded reference. A local winner could lose to a slightly lower-ranked hypothesis when overlap supplied better joint evidence. This was a concrete advance in topology, rather than another attempt to fit uncertain centers. R322, R323

Whole-loop extension exposed implementation and representation limits. R324 processed every region but fragmented into 25 components; 56 links were rejected solely by an artificial chart-boundary rule. R325 removed that fragmentation through cyclic selection, but a changed-context test still contested 174 links. R326 then found that a complete union of individually supported links could lack a simultaneous whole-hypothesis witness in its finite bank. Binary compatibility corrected that defect without proving whole-photo accuracy. R324, R325, R326

R327 made reversible working choices using target-excluded native predictions and common geometry. R328 simplified the inference to connected photographic patches grown from one seed, with alternative histories, overlap checks and bounded rollback. It completed all 187 windows and returned to 17 starting observations with the same lattice translation **H=(214,201)**. The working graph had 900 indexed supports and 1,862 links. R327, R328, growth implementation

That closed *photographic* circuit was a major breakthrough, but not proof of the true string count. Under the two family conventions it implied A=2704 or B=2691. Other retained paths closed differently. R329 traced their first differing row decisions rather than merely comparing final counts. Its later photographic supplement also withdrew an earlier interpretation of a first physical model error at index 31: the frozen model already disagreed with colored material at the reference. R329, corrected diagnostics

October 18 to 19 R330 to R337 A predictive motif and a critical prior correction

R330 applied periodic inference to the photographic lattice rather than insisting that the full optical inversion first be solved. A partial 104 pattern predicted 143 of 296 reserved observations without disagreement, abstaining on the rest. A 208 refinement separated real conflicts between the two 104 halves and explained more of the sequence. Hierarchical prediction, whole-section holdouts and correlated null controls made this much stronger than in-sample visual repetition. R330, periodic inference code

R331 carefully investigated whether 208 was only a spatial alias. The observed A+208 pairs were actual `(16,16)` translations, which also change B by 208; simply renaming the model B/207 could not remove that phase drift. Other maker-compatible candidates were much weaker under the then-hard non-13 rule. Importantly, the report also incorrectly called 299 nondivisible by 13. R337 later corrected that arithmetic: 299=13×23. R331, R337

R332 expanded the working lattice by 141 indexed observations and 423 links, largely by using retained photographic candidates that earlier adapters had omitted. **122 of 126 new inventory observations came from old caches; only four met its strict fresh-extraction definition.** Much of the apparent need for more detection had instead been a failure to exploit existing evidence. Its display audit also found that prior illustrations omitted many already-recorded labels and third-family links. A sparse picture had understated the actual graph. R332

R333 recovered repeated black/red and black/yellow staircase fragments and used pattern breaks to localize possible color or row errors. R334–R336 then investigated pigment and identity separately. In particular, R336 exposed two phantom proposals caused by thresholding dark material near predicted vacancies, moving to the component centroid and resampling it. Those responses were genuine pixels, but not evidence of separate beads. A typed evidence contract and maker decisions corrected their interpretation and prevented admission from darkness alone. Five links formerly blocked by weak/phantom observations were repaired. R333, R334, R336

In R337 the maker changed “not divisible by 13” to **a preference rather than an exclusion**. The final candidate no longer conflicted with that instruction. The spatial and string descriptions then fit together:

```
v = (16,16)       A(v) = 208
u = (6,−7)        A(u) = 0
H = (214,201) = 13v + u
A(H) = 7×214 + 6×201 = 2704 = 13×208
```

The area of the lattice domain generated by v and u is 208. R337 produced 176 strictly supported entries with unresolved contradictions retained, and strict witnesses against every proper divisor of 208. This was the beginning of the final conditional pattern, not an independently measured original count. R337

October 19 to 20 R338 to R344 Simpler pigment evidence and a near-complete candidate

R338 added one already maker-confirmed black body after independent association checks, but did not improve the strict count of supported pattern entries. Its careful vacancy audit rejected same-body shoulders and retained unresolved dark responses. This checked whether “more evidence”

actually meant another independent bead. R338

R339 moved away from increasingly elaborate reflection/annular classification toward simpler body darkness. R340 reduced blackness to a fixed **contiguous 3×3 minimax maximum-RGB measurement, threshold 73.5**, inside a proposed glass region. This had useful interior results while exposing actual region-placement mistakes. A classifier can correctly distinguish black glass when given glass pixels and still fail because its automatic window samples a bright shoulder or a shadow. R339, R340, compact implementation

R341 inspected every active black/red context and controls, making twelve native-supported red-to-black corrections. R342 applied additional explicit maker colors and fixed-pixel sampling, keeping hue separate for red/yellow. R343 then examined **2,377 pixels from 25 inspected glass cores**. It showed that shaded yellow can shift strongly toward orange and that value alone does not determine hue. It also found a boundary-sampling problem and a provenance problem: a general illustration approval had been treated as individual maker-red confirmation. Correcting that scope allowed a useful pigment question without erasing the earlier approval. R341, R342, R343

The strict progression is informative because these late improvements mostly held identity, XY, coordinates and topology fixed:

Round	Indexed observations	Supported / conflicting / insufficient positions
R337	1039	176 / 24 / 8
R338	1040	176 / 24 / 8
R339	1040	179 / 21 / 8
R340	1040	181 / 19 / 8
R341	1040	191 / 9 / 8
R342	1040	195 / 5 / 8
R343	1041	197 / 4 / 7
R344	1041	199 / 2 / 7

The extra indexed observations in R338 and R343 are separately identified admissions, not hidden changes in the evaluation denominator. R344 also reports matched historical cohorts and unchanged-color targets. The improvement is therefore much more informative than comparing favorable subsets from unrelated rounds.

R344 applies two explicit maker-yellow decisions, presents strict and provisional patterns separately, quantifies predictive fit, and restores the original alternating 6/7 graph-paper form. Its side extensions are mathematically correct copies, not extra photographic evidence. The seven-per-side refinements and stepped cuts corrected presentation, without changing the scientific pattern. R344, general graph-paper method

The graph-paper equations make the relationship to crochet explicit: `x=a−b`, `y=a+b`, hence `2k=x+13y` under A. Rightward moves advance k by 1, up-left by 6 and up-right by 7. A thirteen-unit strip contains alternating six- and seven-bead rows; a 208 block spans 32 rows. The same-index seam translation is `(13,−1)` in XY, so a rightward copy moves down one row. This is a verified way to display the candidate's continuity, including its uncertain entries; it does not supply additional photographic observations. General method, presentation geometry implementation

October 20 R345 and R346 A complete scene and a better global shadow

R345 turns the provisional pattern into a complete actual POV-Ray scene. It reuses the original rounded bore-bearing macro, corrected radius, a shared thinner profile, the old reproducible image-axis spline and smooth global placement fields. No later experimental physical fit is silently promoted to truth. All uncertainty in the nine provisional residues remains attached to every emitted bead. Scene generation code, scene provenance

R346 then changes lighting while sealing camera, geometry, pattern, materials and paper. Its first narrow paper-reference band was contaminated by the broad shadow and was rejected. The final reference uses distant paper across the image, enabling a meaningful full-loop comparison. Its success is not a recovery of the bead specular field, and should not be judged by those earlier experiments' unresolved highlight residuals. R346, shadow analysis code

October 20 R347 A finite conditional proof and maker answers

R347 separated a testable pattern claim from physical rendering. It enumerated every divisor below 400 of conditional A=2704 and B=2691, preserving a contradictory-color witness for each rejected case. The original R344 labels rejected every case; a three-yellow hypothetical case admitted only A/208. This was a finite conditional calculation, not an optimization result or proof of its physical premises. A source-linked form asked about three contrary records and seven undercorroborated black entries. Initial proof and question record, finite certificate

The maker answered the first three questions. REC266-P035619 at83 and R332-R-0197-0183 at189 were judged yellow from the original photo. For R332-R-0037-0040 at83, the maker selected yellow but explicitly consulted the reconstruction, said the photographic evidence was inadequate and requested exclusion. The explanatory note changes the evidential meaning of the selection: it is not a third independent yellow confirmation. The maker also reiterated earlier instructions to skip unreadable dark edge cues, which should have prevented presenting this cue again as a mandatory conflict. Exact exported answers

The separate assessment withholds that pigment and applies the two photo-based judgments. Position 83 retains six yellow observations in five geographic regions; 189 has eight in seven. Thus both satisfy the unchanged three-observation/three-region rule. Nothing is majority-recolored, no new observation is created and no pattern color changes. The result is 201/0/7 and 1040/1040 retained agreement; questions 4–10 remain unanswered. Repeating the finite divisor calculation again admits only A/208. The original chart and physical necklace are reportedly available but have not been supplied; their availability is not verification. Assessment and four evidence epochs

R347 retained the cue's working identity and links. R348 tests the stronger hypothetical deletion separately, as described below. This resolves the immediate arithmetic-dependence question without silently treating unreliable pigment, uncertain identity and absent graph constraints as equivalent operations.

4 Technical analysis

Eight coupled problems with different endpoints

Problem	Effective methods and turning points	Final assessment
Identify beads and colors	HSV exploration; positive interiors; native support review; R336 identity safeguards; R339–R343 blackness and actual-glass hue checks	Substantial working inventory and much improved labels. Some source ownership and pigment interpretations remain uncertain.
Locate individual beads	Peaks, connected bright patches, measured interiors, explicit source coordinates	Reliable locations for many *features*. They are not generally physical bead centers or one feature per bead.
Recover arrangement and ordering	Manual graph cycles, joint two-family inference, connected patch growth and closure	Strong working lattice with retained alternatives; no universal independent correctness certificate.
Recover color pattern and continuity	Partial-word/residue constraints, spatial translation, whole-section prediction, strict source support; R347 maker answers and finite divisor witnesses	201 supported positions, seven provisional black entries lacking strict corroboration, no retained color conflict. A/208 uniquely compatible within stated premises; original design and physical continuity independently unverified.
Model 3D crochet geometry	Exact rounded annulus, transported frames, shared dimensions, derived contact and curvature-aware spacing	Final scene physically nonintersecting. Mechanical equilibrium, actual thread path and individual bead poses unmeasured.
Match photo beads to model beads	First-hit ownership, unmatched alternatives, discrete/continuous fitting; later lattice-to-index map	Many plausible correspondences; global mapping remains conditional. Low feature error alone never established identity.
Estimate camera, shape and illumination	Shared-camera fits, dimensional cues, sensitivity audits, common-light trials, final paper-shadow inversion	Camera and dimensions approximate; global cast-shadow lighting materially improved, absolute physical calibration absent.
Render and compare the full photograph	Native POV-Ray scene, same-coordinate crops, exact transforms, paper-shadow profiles	A recognizable and close complete reconstruction, visibly different at enlarged bead scale.

Why a feature is not a bead

The central difficulty was repeatedly an ownership problem, not just a localization problem. A black bead may produce several highlights, a red-looking reflection, or no clear highlight. A colored bead may have disconnected bright lobes. A dark component can be a bore, bead margin, gap or cast shadow. Two same-color bodies can form one connected color region. Consequently, stable peak position, hue agreement, connectivity and small spatial footprint do not independently identify a physical bead.

R193's same-color spill, R231's synthetic controls, R289's multiple red features and R336's phantom responses are different manifestations of this issue. The later separation into source feature, body identity, lattice site, string index and inferred pattern color was essential. It prevented the same image evidence from being counted several times under different names. Ownership audit, seed controls, R289, R336

Why topology could be solved more effectively than optics

The construction supplies strong integer constraints. Three primitive directions form triangles; shared observations in overlapping patches must have the same coordinates; distinct bodies cannot occupy one site; a full circuit must return with a common winding translation. These constraints remain useful when shading, apparent center and exact bead silhouette are poorly measured.

But local cycle consistency is not sufficient: a wrong bridge can pass it, and a whole patch can consistently follow the wrong row. R322/R323 demonstrated this directly. The effective combination was **joint neighboring-row evidence plus exact graph consistency**, not either alone. R328's alternative histories made later context able to revise earlier choices. Its score remained heuristic, so retained coherent alternative closures still matter.

The final index map A=7a+6b has compelling compatibility with spatial `(16,16)` recurrence and H=(214,201). A geometrically consistent B convention exists, but its simple competing repeats predict these colors much less well. That favors A without making all other global photographic interpretations impossible. R323, R328, R331, R337

Why geometry and camera remained underidentified

Absolute bead radius, camera scale and world units have a scale freedom. A planar axis can be deprojected under different camera hypotheses. Thickness, minor radius, phase, tilt, spacing and exposure can compensate for one another. Positive interiors constrain where glass must appear, but not the complete hidden surface or which of several nearby bodies owns a small patch.

R160 supplied an actual ambiguity witness: competing hands fit the positive constraints. R298 exposed a consequential previously fixed camera rotation. R301 showed that a model with excellent highlight locations could be too small to cover the colored material. R319 measured near-duplicate parameter responses and inactive keys. These observations explain why “give the optimizer more parameters” and “hold the old source dimensions fixed” both failed as general prescriptions. R160, R298, R301, R319

The final R345 model does something narrower and useful. It constructs a complete, coherent realization with fixed count, provisional colors, common shape and smooth station/phase adjustments. Its support-proxy residual is **14.43px RMS, 6.58px median, 27.19px P95, 94.73px maximum** over 1,041 supports. These are explicitly approximate bright-support-to-glass-proxy errors, not measured bead-center or silhouette errors. There is no justification for describing the final geometry as recovered at those accuracies. R345 validation

Specular reflection and cast shadow answer different questions

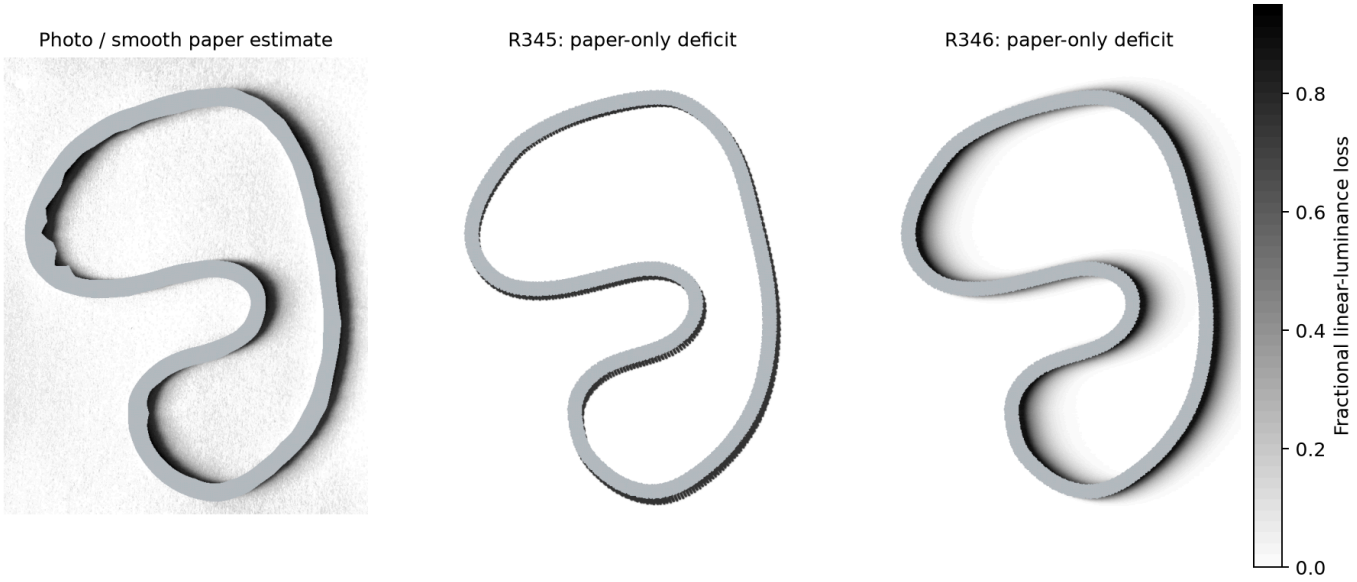
Specular-feature fitting depends on local surface orientation, roughness, illumination, visibility and whether the photographed bright patch is actually specular. Colored/transmitted/diffuse contributions further complicate the target. R276, R289 and R299–R319 show that changing geometry or

correspondence can improve reflection residuals without establishing the correct bodies.

The cast shadow on paper integrates the **whole necklace**, its height above the paper, source position/extent and ambient/direct balance. It is a broader and in this case more tractable lighting cue. R346 evaluates 96 stations on both sides, retaining 192 contours and admitting 142 after grayscale boundary/uncertainty checks. Sample bands exclude photographic material and the exact projected model silhouette. Held stations are every fourth geographic station. R346 method

Whole-necklace shadow measure	Photograph	R345	R346
Median reach to 10% paper loss, same 42 strong photographic profiles	125.5px	27.4px	123.2px
Median outer 90–10% falloff	117.3px, n=42	7.5px, n=24	120.9px, n=41
Paper-profile RMS error, all eligible contours	—	20.4 points	6.8 points
Paper-profile RMS error, held stations	—	20.0 points	6.4 points
Same-coordinate paper RMS error, held stations	—	19.8 points	8.1 points

The differing falloff denominators are important: missing model falloffs are not silently assigned favorable widths. RMS “points” are percentage points of normalized linear-light paper loss, not a percentage of physical correctness.



Gray excludes material and uncertain boundaries. The photographic field uses an estimated smooth paper reference, rather than independently measured unshadowed illumination.

The final source is at `(-5500,-700,4000)` in arbitrary scene units, diameter 6000, with a fixed tilted emitting plane and a shared warm room contribution. Neither source size nor tint varies by section. Individual highlights do not enter the fit. The background estimate is a robust quadratic log-RGB surface fitted to 23,312 distant grid points, with separate residual controls. ICC conversion sensitivity retains the improvement without relighting the scene. These controls support the **improvement**, while paper-reference error, silhouette error, approximate radiometry and coupled height/source geometry limit physical interpretation. Under-body contact darkness is mostly hidden and was not invented. Configuration, R346 measurements and limits

5 Why the investigation took so long

Inherent difficulty

Occlusion and repetition create multiple plausible discrete explanations. One view lacks direct evidence for many conditional addresses. The final map indexes 1,041 of 2,704 assumed positions; the other 1,663 lack indexed observations, which is not the same as proving they are invisible. Repeated colors reduce unique local identifiers. A coherent row switch can preserve many local relationships while changing the winding and downstream pattern phase.

Image formation obscures the relevant physical variables. Surface features are not centers. Hidden bead profiles are unavailable. Camera, material, geometry and lighting compensate for each other. The R160 ambiguity and the optical/material tradeoffs in R301 are direct examples; exact physical recovery from this image cannot be presumed merely because the target exists.

There is no supplied independent full ground truth. The maker reports that the physical necklace and original chart are available but may take time to find. Neither has yet been examined. Known synthetic scenes answer implementation questions, but often fail to reproduce the real photograph's failure modes. Human review of the same image improves evidence but cannot reveal occluded material. Geographic holdouts reduce some local overfitting; they do not create a second independent photograph.

These are real constraints, but they do not account for every delay. The history contains avoidable computational and organizational problems as well.

Methods and implementation choices

Documented issue	Specific evidence	What it explains and what it does not
Wrong surrogate for bead order	April region sorting, background-to-background path and greedy nearest-neighbor tracing	Period inference could fail even with good color masks. It does not imply graph-based ordering itself was a bad idea.
Validation simpler than the actual failure	January toy fixtures; R231 old and new seeds both pass synthetic ownership	Passing tests did not validate the real photographic repair. Useful synthetic tests were retained, but their scope needed to be narrower.

Documented issue	Specific evidence	What it explains and what it does not
Matching aggregate counts instead of identities	R263 brightness cutoff reaches the count target while losing most dark-enclosure supports	A plausible proposal count could conceal worse coverage. It did not validate an expected visible census.
Center-distance/density bias	R214 and R253 high-density controls	More small model bodies can place centers near features while explaining less material.
Legacy dimensions treated too literally	Sine convention; R276 collisions; R278 correction; R321 spacing repair	Several physical failures were model/placement defects. Their repair did not establish correct photographic correspondence.
Fitting a weak or incomplete objective	R301 highlights improve while material omission increases; R307 sparse objective samples; R308 unresolved-ray reward	A lower score was not necessarily a more accurate scene.
Local search and approximate forward ranking	R301 known-synthetic failures; R311 missed basins	Some useful solutions were representable but missed. The history does not establish that one undiscovered global optimum would solve every problem.
Restrictive or inactive parameterization	R298 missing camera rotation; R319 fixed fields, inactive stored parameters and near-collinear responses	Declared parameter lists overstated actual search freedom. More active parameters still required identifiable evidence and physical checks.
Failed extension of local progress	R297 global station normalization and closure-phase change	Earlier successes could regress when the problem was enlarged, even without an intentional change to old local parameters.
Artificial graph fragmentation and incompatible hypotheses	R324 chart gate; R326 finite-bank compatibility audit	Some uncertainty came from the algorithm's representation, not solely from the image.
Existing observations not fully used	R332 recovers 122 of 126 additions from old caches	More sophisticated detection was not always the missing ingredient.
Self-confirming evidence generation	R336 vacancy → dark component → centroid → enlarged response	Native pixels could be real while the proposed new bead was false.

These examples are stronger than a general accusation that optimization was “stuck.” Several have controlled interventions or exact bookkeeping witnesses. Other causes remain hypotheses: for example, unrestricted 3D centerline freedom, a more elaborate camera, or a different glass model might improve the photo, but the history does not show they were necessary or sufficient.

Distinguish the failure before choosing the remedy

Optimizer failure means an admissible good solution exists but the actual search/evaluator misses it. R301 supplies a known-answer witness: accurate replay at the generating parameters has zero positional error, while even known-owner inverse runs fail to recover that geometry. The coarse

predictor adds a separate ranking approximation. The appropriate first response is a small accurate inverse/control test, not a claim that the model cannot represent reality. R311's missed phase/registration basin is another concrete search limitation, although its local repair did not solve the full photograph.

Model or formulation failure means a required behavior is excluded or the objective rewards the wrong behavior. R263's count target cannot distinguish useful bodies from lost dark supports; R324's chart rule creates artificial fragmentation; R326's union can combine mutually incompatible patch hypotheses. R321's spacing repair shows a specific placement-law defect could be corrected. These call for changing the target, representation or constraint, rather than simply increasing iterations. A failed finite nonlinear search does not by itself prove the absence of a solution.

Measurement nonidentifiability means different physical or discrete interpretations satisfy the available checks. R160's competing refitted families and the camera/scale gauge are examples. In that situation the appropriate output is a qualified family of answers or a request for a discriminating observation. R319's coupled/noisy sensitivities complicate search and identification simultaneously: its affine bounds do not prove nonlinear impossibility. All three diagnoses can coexist, and an unexplained stall must remain undiagnosed until a control distinguishes them. Truth replay, R311, R319, detailed decision procedure

An inadequate objective is also distinct from insufficient model capacity: a good physical solution may be representable, yet a correctly minimized count or highlight loss may favor a scientifically wrong answer. Independent task checks must distinguish that target failure from failure to minimize the stated target. These distinctions matter because wider search is a remedy for neither a misleading target nor a genuinely missing observation.

Earlier decision opportunities, with limits on hindsight

Evidence available at the time	Reasonable earlier decision	What remains uncertain
R231: old rejected seeds also pass synthetic ownership	Keep fixtures as implementation checks; add a failure-reproducing real/synthetic ownership control before adopting a repair	Building a representative glass/shadow fixture is difficult; it was not guaranteed to solve segmentation
R263: cutoff238 reaches the count target while losing92/97 dark-enclosure supports	Reject count as a scientific acceptance metric; report distinct-body/reference coverage and abstentions	Dark enclosures themselves are not certified black bodies or a full census
R297: old local fit worsens after extension	Replay old-region station/phase predictions before further fitting; require a preserved-fit baseline	Some local/global constraints genuinely conflict; preservation is not automatically free
R301: exact synthetic truth replays, inverse recovery fails, and highlights improve as material worsens	Block scaling/adoption of that inversion; isolate approximation, owners and search with known-answer tests	The record does not prove a better optimizer would identify real-photo geometry uniquely
R311: archived search misses joint basins and earlier useful starts	Audit actual joint exploration before interpreting an unsuccessful local search physically	Wider search found partial repairs, not a complete solution; finite coverage cannot certify optimality

Evidence available at the time	Reasonable earlier decision	What remains uncertain
R319: serialized fields inactive, active responses coupled/noisy	Audit the evaluator and parameter influence before adding degrees of freedom or interpreting secant bounds	Some variables are only locally negligible; no broad model exclusion follows
R322–R328: joint rows outperform isolated bands; overlap and compatible patch histories provide structure	Use the manual chart and discrete alternatives as the main pattern route, leaving full optics separate	The whole-loop breakthrough depended on accumulated candidate caches and overlapping context; an equally successful early implementation is unproved
R332:122/126 additions were cached rather than freshly measured	Inspect source/alias/association coverage before designing another detector	New local fitting and careful native review were still needed to use those records
R336: native dark pixels support a false body through prediction and recentering	Require independent body ownership before admitting counting/obstruction/indexing constraints	Weak evidence can still describe real bodies; a blanket dark-edge rejection would lose information
R339–R343: simpler material tests repair some cases, yet window ownership failures persist	Compare a simple owned-glass baseline first; use explicit small material annotations to test sampling	R339 does not dominate all matched controls; a simple scalar or fixed offset is not universally reliable

These decisions were feasible diagnostic changes using then-available evidence. The strongest avoidable pattern was continuing to optimize a surrogate or expand a model before resolving a demonstrated prerequisite failure. The claim that modest annotation plus graph algorithms would have shortened the whole project is plausible and supported by later mechanisms, but remains a counterfactual. It should not be expressed as a promised number of saved rounds or a guarantee that every uncertainty was easy.

Conduct, diagnosis and communication

The objective sometimes became a difficult intermediate product. Broad masks, exact brightness paths, full contours and reflection minimax all received substantial attention before the topology was reliable. The June plan and R239 already allowed a useful partial subset to support the next step; R293 later had to restore that priority. This suggests a sequencing problem, although the final graph did benefit from the accumulated source candidates and surface tools.

Evidence preservation did not automatically preserve scientific progress. The late project carefully sealed old products and retained failed versions. R297 nevertheless lost an earlier local match because its new normalization did not preserve the old physical metric. A preserved file is not the same as a live regression constraint or an explicit decision about an acceptable tradeoff.

The visible explanation sometimes lagged behind the data. R332 found that prior figures had omitted many already-recorded supports and third-family links. R329's corrected same-diagonal diagnostics made a row-choice disagreement understandable through native colored material; earlier model-index comparisons had overstated the first physical error. These are direct examples of diagnostics exposing a discrepancy before presenting a sufficiently clear cause. R332 display audit, R329 correction

Approval scope was occasionally overgeneralized. R343 found that general approval of labels/adjacency had become a protected individual pigment fact. That protected a red interpretation and affected its use as a hue reference until the scope was corrected. This was neither an inherent ambiguity nor a failed optimizer; it was a provenance/communication defect. The later separation of explicit pigment judgments, illustration approvals and software labels is a substantial improvement. R343 authority analysis

Hard prior information needed explicit status and periodic review. Excluding repeats divisible by 13 created a genuine conflict with the best predictive structure. It was legitimate to respect the maker's instruction while it was firm. Once clarified as a hope, it was correct to remove the filter. The lesson is not to ignore maker knowledge; it is to distinguish construction facts, strong recollections, preferences and provisional hypotheses. The arithmetic mistake about 299 was independently avoidable. R331, R337

Small rounds improved accountability but could fragment the objective. Frequent stops, extensive handoffs, tests, review pages and preservation work helped prevent silent changes. The chronology also includes many clarification and presentation rounds between scientific steps. The record supports saying that the process had coordination costs; it does **not** support assigning a numerical fraction of elapsed effort to documentation, reviews or any particular model/session choice.

Abstention instructions needed to become operational evidence policy. The maker had repeatedly asked to skip especially ambiguous dark edge cases. R347 still presented Q1 as a contrary observation to settle, despite insufficient photographic evidence. Its exclusion loses neither residue coverage nor saved-graph connectivity. The useful policy is role-specific abstention before checking motif agreement: unknown pigment must not be forced into a color, and uncertain identity must not silently become a counting or obstruction fact. It is equally important not to discard genuine contrary observations merely because they disagree. Maker note, sensitivity and systematic policy

Finally, the calendar is misleading as a measure of continuous work. There are bursts in April 2025, January 2026, June 2026, late September and October 2026, with substantial gaps. The repository's 2017 origin is the preexisting forward-rendering history. It would be wrong to describe this as nine years of continuous inverse reconstruction or to equate 346 request numbers with 346 independent optimization campaigns.

6 What the final state establishes

Pattern evidence

The **historical R344** strict block contains **62 red, 60 yellow, 77 black and 9 unknown** entries. Its provisional completion contains **62 red, 62 yellow and 84 black**. Its nine then-tentative positions were:

Status	Positions	Provisional choice
Conflicting	83, 189	Yellow at both
Insufficient geographic support	43, 75, 82, 89, 140, 147, 186	Black at all seven

The three historical mismatches and their R347 dispositions are:

Observation	Residue	Historical / provisional	Native XY	R347 disposition
R332-R-0037-0040	83	Red / yellow	2301.35, 1430.23	Pigment withheld; photo inadequate, selected yellow reconstruction-dependent
REC266-P035619	83	Red / yellow	2124.72, 736.69	Maker photo-based yellow correction
R332-R-0197-0183	189	Red / yellow	599.64, 624.45	Maker photo-based yellow correction

They were inherited software-red records, not contrary explicit individual maker pigments. R347's assessment now has **201 supported / 0 conflicting / 7 insufficient**, with **1040/1040** retained colors agreeing and all 208 residues observed. The complete provisional word is unchanged. The seven black positions still lack the old three-region corroboration, and questions 4–10 have not been answered. This is measured agreement with revised evidence, not1041/1041 agreement or independent physical accuracy. Historical native evidence, updated assessment

The historical predictive checks remain useful and are preserved unchanged. R347 did not rerun this cross-validation or turn its two targeted photo judgments into a blind validation set:

Evaluation against historical R344 recorded labels	Agreement
Provisional completion fitted to all indexed observations	1038/1041
Strict entries on observations where the strict block emits a color	1011/1011, with 30 abstentions
Flat 208, geographic block and both neighbors withheld	1024/1041
Flat 208, complete conditional occurrence withheld	1038/1041
208 prediction with a 104-parent hierarchy, guarded geography	1029/1041
Frequency baseline, guarded geography	392/1041
Exact 104 predictor, guarded geography	927/1041

The strict 100% follows its definition of uncontradicted support; it is not independent accuracy. Geographic prediction has 38 targets without a remaining same-residue training observation, 27 of which succeed through the black frequency fallback. Thus high black predictive agreement does not independently measure every tentative black entry. The hierarchy's better hard accuracy and log score also show why one should not report only the most favorable metric. Evaluation data, evaluation implementation

Every proper divisor of 208 has a pair of conflicting **strict** witness colors. Therefore every cyclic completion of the strict block has minimal cyclic period 208 **under the current indexing**. That mathematical statement is solid and useful. It does not prove physical count, exact repetition around

the actual necklace, or that the working indexing is unique.

The preserved active graph inventory is **1,059 observations, 1,041 indexed sites and 2,290 working links**, one component with H=(214,201). R347 uses only 1040 indexed pigments. These are observations and working relations, not a certified complete physical census. R344's 37 explicit individual indexed pigment judgments agreed with the provisional pattern; R347 adds two targeted photo-based judgments in its separate assessment. These selected same-photo judgments are not a random independent validation sample.

Conditional uniqueness and sensitivity to the ambiguous identity

R347 tests every shortest-period divisor below 400 of A=2704 and B=2691 under their specified indexing maps. The two photo-based corrections plus Q1 pigment abstention leave only A/208, with all 208 residue colors determined by retained observations. This extends the proper-divisor primitivity result into a finite uniqueness result **within those premises**. It excludes neither other photographic row histories nor different construction/count assumptions; exact complete repetition is itself a premise. All cases and contradictory witnesses

Color withholding retains the questionable body's proposed occupancy and links. Identity uncertainty instead requires marking those relationships tentative, and full hypothetical deletion removes them. R348 tests the last operation on derived data:1040 nodes and 2287 links remain in one component, with 1248 independent graph cycles rather than 1250. Integrating saved primitive edge deltas, without supplying H, recovers the same cycle-displacement subgroup generator (214,201) and the same remaining coordinates modulo H. The three old neighbors have target-free connecting paths of 1/2/1 edges. A target-free winding cycle is saved as an exact witness. The target also postdates R328's selection of H. Sensitivity methods, results and certificate

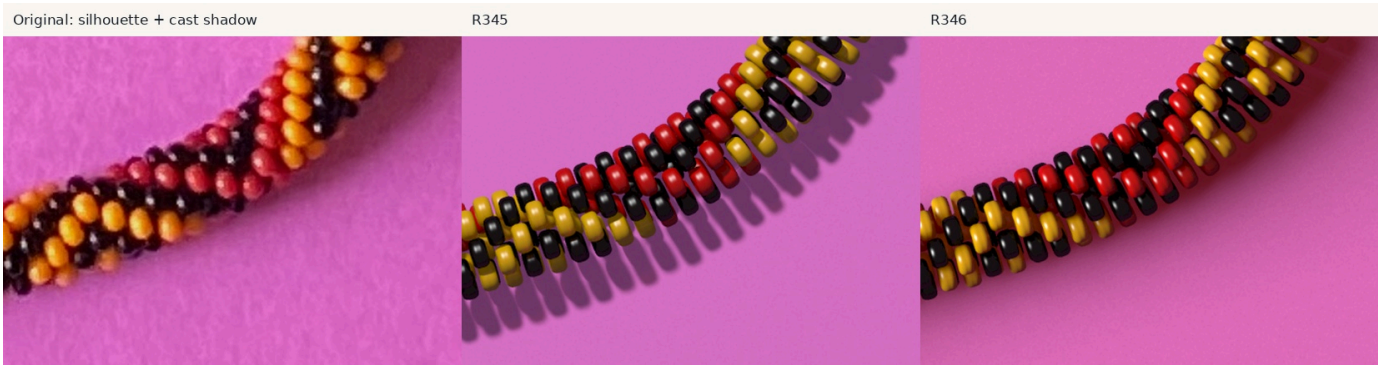
Thus neither the retained graph's lattice/closure constraints nor the conditional A/208 color result depends on keeping this node. This is a statement about the saved graph with other edge labels fixed. R332 used local anchor/occupancy searches; static deletion does not prove that a counterfactual image pipeline would make every same selection. No saved accepted addition names Q1 as a direct fitting anchor, but some provisional/rejected candidates do. The smallest further test, if needed, is local reassessment of affected saved candidates with Q1 quarantined and unrelated inputs frozen, not a whole reconstruction campaign. It has not been run. Physical topology and all-bead correspondence remain independently unverified.

Geometry and rendering evidence

R345's complete scene has common bead radius 1.78453698, shared axial thickness/face-diameter ratio 0.55, a planar carrier, 416 signed minor turns and smooth periodic station/phase fields. It uses the historical image-axis spline rather than adopting R298–R320 as recovered physical geometry. The orthographic camera elevation is 80 degrees and scale 8.5 native pixels per arbitrary world unit. Those are explicit reconstruction choices. R345 configuration, R345 provenance

The nonintersection result is stronger than a sample that happened not to hit overlap. Bounding geometry excludes distant pairs, and positive separating-plane margins for the filled rounded envelopes certify every remaining nearby pair. The smallest margin is about 0.00012883 arbitrary world units; exact minimum glass height is about 0.03000068 above paper. Independent verification

checks emitted transforms, the original macro's bore/surface and camera marker projection. These certify the implementation and chosen scene's geometric properties, not photographic truth.
Independent verification



The shadow becomes much closer in R346. The bead-scale differences remain: regular rounded cylinders, repeated poses, visible spacing, gloss and local color arrangement differ from the photographed relaxed crochet.

Four levels of accomplishment

Standard	Assessment
A plausible reconstruction	**Achieved.** The final scene is coherent, complete, recognizable, executable and nonintersecting.
Fits substantial portions of the photograph	**Achieved.** Many local patterns, the full loop arrangement and the broad cast shadow are close; explicit measurements support more than a visual impression.
Exact bead sequence and topology convincingly established	**Strong conditional evidence, not complete establishment.** R347 resolves the retained color conflicts, supports201 positions and uniquely admits A208 within fixed premises; seven provisional black positions and coherent global mapping alternatives remain. The authentic chart, physical count, topology and correspondence have not been independently checked.
Physically and visually convincing reproduction of the entire photograph	**Partially achieved.** Convincing at overall layout/pattern/shadow scale; enlarged geometry, correspondence, paper texture and optics still differ. No whole-image photorealistic or calibrated physical recovery has been demonstrated.

This assessment does not carry forward obsolete failures. The final rendered scene's interpenetration is solved; the broad cast shadow is substantially improved; black identification is no longer restricted to the early unreliable highlight/hue machinery; the 13-divisibility exclusion is withdrawn; and most late pattern conflicts have actually been resolved. Remaining uncertainty is attached to the latest evidence, not inferred from an early failed round.

7 Lessons and practical alternatives in hindsight

Start with a small trusted photographic graph and a partial color word

Once the maker supplied the 1/6/7 rule, it was practical to annotate a few compact patches with stable IDs, explicit neighbor directions and a small genuine-glass pigment sample. The later 27-bead manual chart shows this was feasible. Exact cycle/duplicate checks could then validate bookkeeping, while source photographs validate the edges. This would have provided a real-image baseline for automatic indexing without demanding broad masks or a solved camera.

The caveat is important: a single patch cannot establish a global winding or all hidden colors. Several geographically separated patches and explicit unresolved bridges would still be required. The alternative is not “do it all manually”; it is to obtain enough independent structure to diagnose an automatic method early.

Some late discoveries depended on real prerequisites. R328 needed compatible overlapping row hypotheses and a sufficiently rich observation bank to test a photographic return. R347's finite uniqueness check needed a coherent lattice/closure and usable color evidence; its result could not simply have been asserted at the start. Quantitative R346 shadow fitting benefited from a complete nonintersecting scene and silhouette exclusions, even though a qualitative whole-shadow check was feasible much earlier. The proposal is to expose and test these dependencies sooner, not assume that every final result was available without them.

Preserve feature types and identity before optimizing them

A source record should distinguish a color sample, bright patch, reflection, body identity, lattice site and pattern prediction from the start. Multiple features on one body should be allowed without silently counting multiple beads. A dark region should never create a new body simply because a predicted lattice vacancy is nearby.

This was practical as a data-design choice before sophisticated imaging algorithms existed. R336's typed contract and R343's approval strata show what the earlier design needed. It would have reduced phantom proposals, false calibration labels and repeated review of the wrong question.

Use topology and recurrence to diagnose optics before solving optics

The most productive late order was photographic graph → candidate string mapping → predictive motif → independent pigment/identity checks → render. That allowed an imperfect but useful topology to reveal color mistakes. Requiring a globally convincing physical fit before any pattern inference consumed effort on underidentified optical targets.

A constrained graph or factor-graph formulation would have been practical after the local manual chart: discrete family/row choices, reciprocal neighbors, overlapping observations and winding consistency, with a small bank of local photographic hypotheses. R323–R328 demonstrate the relevant mechanisms. A monolithic unconstrained joint pixel optimizer would not automatically have solved this; R326 shows that even a global solver can pool incompatible evidence if its representation is wrong.

Pattern predictions should remain separate from source labels. Their useful role is to locate informative disagreements and compare coherent hypotheses, not recolor every minority observation. The final late rounds follow that discipline substantially better than a free edit-distance alignment would.

Audit physical source assumptions before fitting a large scene

The sine units, actual axial height, minor radius, turn advance and hole-axis convention were all accessible from the original source. A compact dimensional audit and a straight/curved packing check at the outset would have prevented treating some rendering defaults as measured construction facts. R278, R303 and R320/R321 demonstrate these calculations were tractable.

This would not have recovered the real geometry by itself. The actual beads need not have the original macro's dimensions, and crochet relaxation is not mechanical equilibrium in that model. The practical gain would have been to distinguish source-parity tests from physical hypotheses much earlier.

Require inverse recovery on representative synthetic controls

Before long searches, use an accurate forward scene with known parameters, perturb the initialization, and check whether the inverse search recovers both geometry and correspondence. Compare cheap and accurate predictors on the same proposals. Include diffuse colored lobes, several highlights, shadowed glass, near-tangent surfaces and same-color touching beads.

This was practical because POV-Ray fixtures and exact geometry were available early. R301's known-answer failure should have been a gating diagnostic earlier in the global-fitting campaign. Yet R231 warns against treating convenient synthetic scenes as sufficient: if the old broken method also passes, the control is not representative of the failure being repaired.

Inspect actual search freedom and preserve useful local fits

For each fitted parameter, record whether it actually changes the active model, its units, tested range, coupling and start coverage. Test periodic variables as periodic, and distinguish numerical bounds from physical constraints. R298 and R319 show why an exported parameter list is not an adequate search-space audit.

When extending a local fit, verify its actual projected surfaces and physical station map before and after extension. R297's metric regression was detectable directly. Maintain component and geographic checks so that an aggregate loss cannot quietly sacrifice a difficult region. These measures were practical with the existing saved supports; they did not require a new detector.

Simplify pigment questions and use manual information where it is strongest

For the known red/yellow/black target, a small verified glass patch and a simple blackness test were more informative than reflection hue, large neighborhood means or complex annular voting. Separate red/yellow hue analysis then needed genuine glass ownership and shadow examples. R340–R343 show both the strength and limits of this approach.

The practical earlier intervention would have been a modest stratified set of explicit body/pigment labels, including dark red, shaded yellow and reflection-free black, with exact pixel samples. That could validate automatic outputs and expose sampling errors. It would not justify assuming the 73.5

threshold or a fixed window generalizes to every photograph; even the final simple rule retains ownership failures.

Use the global shadow as an early lighting cue and keep its baseline independent

The broad shadow was visible from the beginning and is a useful global constraint on source direction, angular size and direct/fill balance. An early paper-profile comparison could have rejected the short sharp shadow of the final R345 baseline before extensive local highlight fitting. The R346 method demonstrates the value of that cue.

However, a quantitatively credible fit genuinely benefited from later nonintersecting full geometry, exact silhouette masks and reliable paper exclusions. Its rejected local reference also shows why “measure the shadow” was not a trivial shortcut. A qualitative global shadow check was practical early; the final rigorous relative-photometry comparison required more infrastructure.

Organize work around unresolved claims

Keep one compact table of claims—body identity, topology, pattern, geometry, camera, light—with supporting evidence, counterexamples and the next discriminating observation. Retain failed epochs, but make the current selected evidence and superseded assumptions easy to find. Review pages should show all relevant observed neighbors, the actual physical question, and what different answers would change.

The existing reproduction and preservation machinery is worth keeping. The improvement would be to make it serve cumulative scientific decisions, rather than allowing each local algorithm or review page to become a new stopping point. The history does not justify abandoning incremental work; it supports larger coherent milestones with clear evidence gates and fewer redundant prerequisite refinements.

The remaining exact-pattern question is now narrower than the R344 recommendation: the two contrary photo labels have been corrected, and the unreadable third pigment has been withheld without loss of coverage. The seven provisionally black residues and physical/global-mapping premises still need evidence appropriate to an exact claim. An authentic original chart is a direct check of the intended design under cyclic offset and reversal; it does not prove every assembled bead follows it. Re-questioning the unreadable Q1 as though it were a necessary conflict would repeat the process error.

Highest-priority changes for the next project

The strongest lessons are to separate pattern recovery from full optics; establish a small trustworthy real-photo graph early; gate inverse campaigns on known-answer recovery and representative failure controls; and preserve independent task constraints when scores improve or fits expand. Active-parameter/search audits and a concise claim register should make these gates operational. Simple owned-material measurements deserve an early matched baseline, while their ownership errors remain explicit. More ambitious robust joint graph/period solvers and additional-view acquisition are promising, but their total benefit has not been measured here.

An early warning should trigger a decision, not merely another report: stop adoption when essential correctness regresses, stop scaling when known-answer recovery fails, reconsider when the same independent counterexample survives repeated refinements, and seek new evidence when several coherent interpretations remain equivalent. Passing software tests and producing more artifacts do not override these conditions. The detailed future-project methods gives costs, effectiveness checks, ranked recommendations, separate bead-pattern/rendering workflows, twelve failure diagnostics and a reusable `AGENTS.md` checklist. No new scientific campaign is begun by this recommendation.

Evidence links

The reference definitions below use immutable Git snapshots. Relative links to later R347/R348 documents supplement them. Reports are paired throughout with implementations, numerical artifacts or original images where their distinction affects the conclusion. Existing receipts are historical artifacts: the original retrospective's static recount reproduced R344 199/2/7 and three mismatches, the emitted scene data reproduced 2,704 exact pattern assignments, and saved profile arrays reproduced the R346 RMS comparison. The original document/preservation checks, including its original report hash, are preserved as an earlier snapshot, not a receipt for this updated text. R348 adds saved-graph analysis and current validation. No historical optimizer, detector or renderer was rerun.